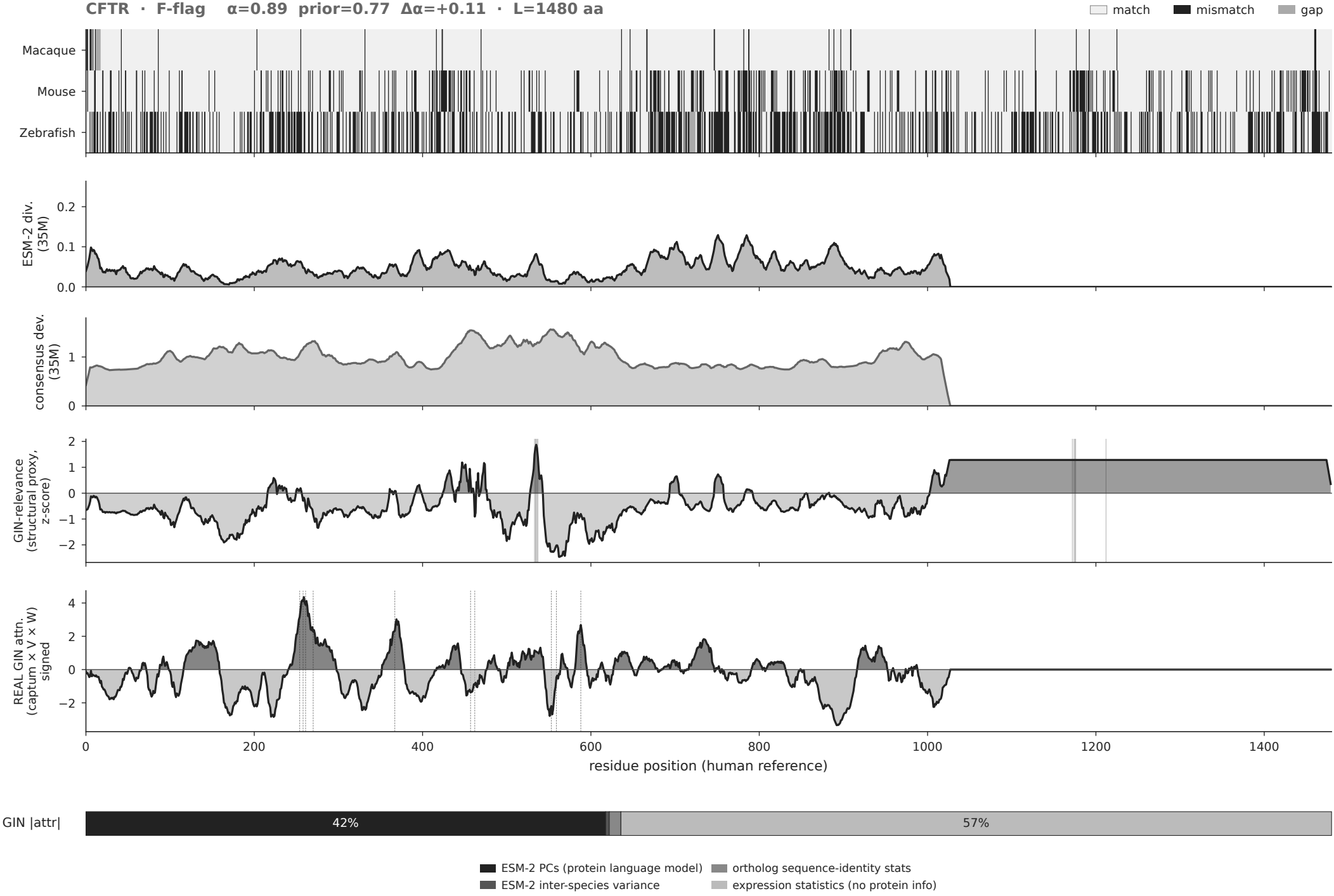

CFTR · F-flag α=0.89 prior=0.77 Δα=+0.11 · L=1480 aa
match
mismatch
gap
Macaque
Mouse
Zebrafish
ESM-2 div. (35M)
0.0
0.1
0.2
consensus dev. (35M)
0
1
GIN-relevance (structural proxy, z-score)
2
1
0
−1
−2
REAL GIN attn. (captum × V × W) signed
4
2
0
−2
0
200
400
600
800
1000
1200
1400
residue position (human reference)
GIN |attr|
42%
57%
ESM-2 PCs (protein language model)
ESM-2 inter-species variance
ortholog sequence-identity stats
expression statistics (no protein info)